Types of Use Cases in Software Engineering

1 Introduction

Use cases can vary in their purpose, level of detail, and perspective. To effectively describe system behavior, software engineers classify use cases into distinct types. This classification helps organize requirements, determine the level of abstraction, and identify stakeholders' goals.

Definition

The **type of a use case** indicates its scope and level of detail—whether it describes high-level business objectives or specific interactions between a user and the system.

2 Classification of Use Cases

Use cases are generally categorized in three main ways:

- Based on **scope** — what level of the system they describe.
- Based on **detail** — how much information they include.
- Based on **goal abstraction** — whether they describe business or system-level processes.

3 1. Business Use Case (High-Level)

A **Business Use Case (BUC)** represents the interactions between external stakeholders and the organization or business process. It focuses on what the organization does, not on the internal software or system components.

Key Idea

Business use cases model the business goals and workflows of an organization before any system design begins.

Characteristics:

- Describes business processes rather than technical details.
- Focuses on **actors** such as customers, suppliers, or departments.

- Used during the **business modeling phase**.
- Often represented in business process diagrams or domain-level models.

Example:

Business Use Case Example – Online Order Fulfillment

Actor: Customer
Goal: To purchase and receive a product online.
Description: The customer browses products, places an order, makes payment, and the organization processes and delivers the product.

4 2. System Use Case (Technical-Level)

A **System Use Case (SUC)** describes the interaction between a user (actor) and the software system. It focuses on how the system behaves to support a business process.

Key Idea

System use cases translate business requirements into technical functionality that developers can implement.

Characteristics:

- Focuses on how the system supports user goals.
- Describes step-by-step interactions between actor and system.
- Written in structured text or UML diagrams.
- Created during the **system analysis phase**.

Example:

System Use Case Example – Process Online Payment

Actor: Customer
Goal: To make a secure online payment.

Main Flow:

1. User enters payment details.
2. System validates payment information.
3. Payment gateway authorizes the transaction.
4. Confirmation is displayed to the user.

Postcondition: Payment is successfully recorded in the system database.

5 3. Essential vs. Real Use Cases

Use cases can also be divided by their level of detail and abstraction.

5.1 Essential Use Case

An **Essential Use Case** describes what the user wants to achieve and what the system must do—**without specifying implementation details**.

Example – Essential Use Case: Submit Feedback

Goal: User provides feedback to the system.

Steps:

1. User chooses to submit feedback.
2. System prompts for input.
3. User provides comments.
4. System records feedback.

Note: No mention of UI design or storage method.

5.2 Real Use Case

A **Real Use Case** includes implementation details such as interface elements, system responses, and data handling steps.

Example – Real Use Case: Submit Feedback

Goal: User submits feedback through a web form.

Steps:

1. User clicks "Feedback" on the navigation bar.
2. System opens a feedback form with text fields and rating options.
3. User enters comments and clicks "Submit".
4. System stores feedback in a database and displays a thank-you message.

Note: Includes technical and UI details.

6 4. Summary Table of Use Case Types

Type	Focus	Purpose / Example
Business Use Case	Organizational or business goals	Order processing, service request handling
System Use Case	Interaction between actor and software system	User login, payment processing
Essential Use Case	Conceptual level, no implementation details	Describes goals and outcomes
Real Use Case	Concrete level, includes UI and system responses	Includes forms, screens, data storage

7 Summary

Understanding different **types of use cases** helps software engineers model systems at appropriate abstraction levels. High-level business use cases capture organizational goals, while detailed system use cases define how software fulfills those goals. Essential and real use cases ensure a smooth transition from concept to implementation.

Each type of use case tells the same story from a different perspective — from the business vision down to user-system interaction.